Shooting Guide – one spot, one speed

The robot's flywheel and windmill are now consistent (robot logs 30 Oct: every ball left at 1200–1260 t/s). Misses come from **distance and aim**: one speed only lands from one distance. So we tune **one shooting spot** and always shoot from it.

0. Before you start

- Practice HIVE with the **AprilTags on the bottom of the CELLs** (IDs 30–45, printed for practice from the FIRST field resources page). Without them the camera has nothing to see.
- Pick the **alliance that matches the HIVE you shoot at**. AUTO RED / AUTO BLUE sets it on its own; in TeleOp INIT press B / Circle = RED, A / Cross = BLUE. The robot remembers it (even after a restart). **Always read the "Alliance" line in INIT before PLAY** – the camera only looks for that alliance's CELLs.
- Fresh battery (12.5 V or more), 4 balls, tape measure, coloured tape.

1. Pick ONE shooting spot and tape it

1. Choose a spot in front of your alliance's HIVE that drivers can reach quickly and that is not in a lane the other alliance drives through.
2. With the printed hood (Nov 2026): shoot **straight in front of the CELL, robot front about 2 in from the field wall** (the ball leaves about 3 ft from the CELL opening; with one flywheel wheel the camera picks about 1440–1465 t/s at 28–44 in). The camera sees the tags better when not too far.
3. Put tape on the floor where the **front edge and both sides of the robot** go (an "L" at two corners works well), robot square to the HIVE.
4. Write down where the spot is using things that are the same on every field (tile seams count from the walls, distance from the HIVE frame):

	Our spot (fill in)
Alliance / HIVE side	
Tiles from our alliance wall (count seams)	
Tiles from the audience / rear wall	
Floor distance, camera lens → point under the CELL tags	in
Robot heading (which wall is behind the launcher)	

2. Find the speed: BioBuzz Shot Log

Robot on the tape, square to the HIVE. Select BioBuzz Shot Log, press INIT, then PLAY. D-pad = the + shaped arrows on the left of the controller; in Shot Log it changes the speed being tested.

Button	What it does
Y / Triangle	Flywheel ON / OFF (starts at the NEAR preset, 1450 t/s)
D-pad UP / DOWN	Speed +50 / −50 t/s (big steps)
D-pad RIGHT / LEFT	Speed +10 / −10 t/s (fine steps)
RB (right bumper)	Fire ONE ball (only when the flywheel is at speed)
Then A / Cross	Ball went IN the CELL
Then B / Circle	MISS
Then Back / Share	Discard (no ball came out)
Triggers	Intake in / out to reload

1. Flywheel ON (Y). Fire 10 balls at **1450**, marking each one IN or MISS.
2. Mostly **SHORT** (hits the front / falls before the CELL) → D-pad UP (+50). Mostly **LONG** (over the back) → D-pad DOWN (−50).
3. When it is close, use RIGHT / LEFT (+/−10) and repeat 10 balls.
4. The Driver Hub shows the hit rate for each speed. Keep the speed with **8 or more of 10 IN**.

CHECK: Do not move the robot between shots – re-check it is on the tape after reloading. Left/right misses are aim, not speed.

Speed (t/s)	IN	MISS short	MISS long	MISS left/right
1450				

DONE WHEN: One speed gives 8/10 or better from the tape. Final speed: _____ t/s

3. Put the speed in the robot

- Tell the programmer (or Copilot) the speed. In **BioBuzzConstants.java**:
- `AUTO_SPEED_DIST_IN / AUTO_SPEED_TPS` = the TeleOp auto speed table: put your speed at your spot's camera distance (the "HIVE aim" line shows it).
- `LAUNCHER_NEAR_VELOCITY` = your speed (used when the camera can't see the CELL, and by D-pad DOWN = NEAR).
- `AUTO_LAUNCH_VELOCITY` = the same, only if AUTO shoots from about the same distance.
- Rebuild, deploy, then fire 5 balls in TeleOp from the tape to confirm.

3b. Save the SHOT SPOT on the robot (Share button, no code change)

1. Start BioBuzz TeleOp (check the alliance in INIT). Put the robot on the tape, launcher facing the CELL.
2. Check the “Shot speed” line: it says AUTO and the camera picked the speed. Use RIGHT / LEFT until it shows your Shot Log speed.
3. Hold X until the bearing settles, release, then press **Back / Share**. The controller rumbles and the screen shows “SHOT SPOT saved: 40 in to BLUE_REAR @ 1440 t/s” (the distance, the CELL it belongs to, the speed).
4. Fire 4 balls to confirm. Not good? Move, re-tune the speed and press Share again – the new save replaces the old one.

RESULT: From now on, **holding X or RB** turns the robot to the CELL and drives it forward/back to the saved distance (only while that same CELL, or the same CELL on the other alliance’s HIVE, is in view); the camera sets the speed for that distance. At the spot the controller rumbles and the screen shows **ON SPOT - FIRE**. One save works on both alliances and survives restarts. AUTO shoots at the saved speed.

CHECK: Share only saves when the camera sees your alliance’s CELL tags (otherwise: “Can’t save the spot”). Pushing the left stick forward/back while holding X takes over the driving. A spot saved before 10 Nov night shows “? - press Share again”: X / RB only aim until you save it again.

3c. Teach the AUTO speeds in practice (LB + D-pad)

In BioBuzz TeleOp the robot logs every ball it launches: the camera distance, the speed, where the robot was. Tell it what each ball did, and the laptop works out the right speed for every distance (no more trimming by hand).

1. In INIT, check the **Diagnostics log says ON** (it is on by default; Back / Share toggles it).
2. Shoot from different distances; the camera sets the speed (AUTO). After each ball press **LB + D-pad**: UP = IN, DOWN = SHORT (too slow), RIGHT = LONG (too fast), LEFT = SIDE (left / right / bounced out: aim, not speed).
3. A buzz = saved. Two blips = no ball waiting (all labelled, or more than 10 s ago). A scout can call the results out; for a volley, label the balls in the order they flew. The “Balls” line shows the counts.
4. After practice, on the laptop: pull the logs (`tools\pull_robot_logs.ps1`), then run `python tools/shot_results.py robot_logs <date>`. It shows every table distance with its IN / SHORT / LONG balls and proposes new speeds (a distance moves only with 3 or more labelled balls near it). `--write` puts them into BioBuzzConstants.java; then rebuild and install.

CHECK: Many SIDE balls = aim, not speed: check the aim trim in BioBuzz Tune. Use a good battery (12.5 V or more): a weak battery makes balls short and teaches the wrong speeds.

DONE WHEN: At every distance you shoot from, most labels are IN and the tool proposes no more changes.

4. Camera and AIM check: BioBuzz Vision Test

1. Robot on the tape. Select BioBuzz Vision Test, INIT, PLAY. B / Circle switches alliance – pick the HIVE you face.
2. The screen should show your CELL with 2–4 tags and a bearing near 0. Hold X / Square: the robot turns until the bearing settles.
3. In TeleOp the line “HIVE aim” now always shows what the camera sees (cell, tags, bearing, range) – even when you are not pressing X.

CHECK: “no RED CELL tags in view (any tags: 3)” means tags are seen but they belong to the other alliance’s CELL – fix the alliance in INIT. “any tags: 0” means the camera sees no tags at all: check the practice tags, the camera tilt (30° up) and that nothing blocks the view.

DONE WHEN: Your CELL shows with 2+ tags from the tape, and holding X turns the robot onto it.

5. DRIVER RULE (match play)

1. **DRIVE NEAR THE SPOT** – launcher toward our CELL, press Y to pre-spin.
2. **HOLD RB** – the robot turns onto the CELL and drives to the saved distance by itself.
3. **WAIT FOR THE RUMBLE** – screen shows ON SPOT - FIRE.
4. **KEEP HOLDING RB** – it fires when aimed, one ball at a time.
5. **RELEASE** RB when empty, then drive away.

No tags seen (no rumble)? Drive onto the taped spot by eye, square up, then RB. Do NOT shoot while moving.

Tip: press Y (pre-spin) while driving to the spot, so the flywheel is already at speed when you stop.

TeleOp button card

D-pad = the + shaped arrow buttons on the LEFT side of the controller. In TeleOp it changes the SHOT SPEED (how hard the ball is thrown); with LB held it saves what a ball did, for the log. It does not drive or aim. Normally you don't need UP / DOWN: the camera sets the speed from the distance (AUTO).

D-pad (TeleOp)	What it does	Use it when
UP	MANUAL FAR: 1570 t/s, trim reset to 0. The camera speed is off until you press X	camera can't see our CELL, far away
DOWN	MANUAL NEAR: 1450 t/s (or the speed you tuned), trim reset to 0. The camera speed is off until you press X	camera can't see our CELL, near
RIGHT	+10 t/s (each press), stays on top of AUTO	balls land SHORT (in front of / below the CELL)
LEFT	−10 t/s (each press), stays on top of AUTO	balls go LONG (over / behind the CELL)
LB + UP / DOWN / RIGHT / LEFT	What the ball did, saved in the diagnostics log: UP = IN, DOWN = SHORT, RIGHT = LONG, LEFT = SIDE (left / right / bounced out). Goes to the oldest ball of the last 10 s without a result: label a volley in order	practice: after every ball (a scout can do it). tools/shot_results.py turns the labels into new AUTO speeds

The Driver Hub line "Shot speed" always shows it, e.g. `AUTO 41 in 1445 +10 = 1455 t/s`. AUTO keeps the last speed for 3 s when the CELL drops out of view, then uses the last preset (NEAR at the start); while RB is held it follows the distance until the first ball goes, then stays until RB is let go. Pressing UP or DOWN clears the +/− trim and switches to MANUAL (RB keeps it); X switches back to AUTO.

CHECK: D-pad in other programs is different: BioBuzz Shot Log = speed ±50 (UP/DOWN) and ±10 (RIGHT/LEFT); BioBuzz Calibration and Arm Setup = arm position; AUTO INIT = start delay.

Button	Action
Left stick / right stick X	Drive + strafe / turn. ROBOT-CENTRIC by default: stick up = intake side. FIELD-CENTRIC is optional (L1 in INIT, or hold Options 1 s): stick up = away from you, whichever way the robot faces. While RB or X aims at a CELL the stick is robot-centric
Options	Hold 1 s: field-centric on / off (1 blip = on, 2 = off). Tap (field-centric only): point the intake away from you and tap it after a big hit; the start comes from AUTO, or the robot is assumed to have its intake pointing away at INIT
LB (hold)	Precision (slow) drive
RT / LT	Intake IN / OUT (half LT = flower back-out)
RB (hold)	AIM + FIRE: spins up; with our CELL in view the robot turns onto it and drives to the saved SHOT SPOT distance by itself, then feeds one ball at a time when the flywheel is steady and the aim is good. Without the CELL it fires as before. FIRE HELD on the screen says why; the ball goes anyway after 1.5 s. Your D-pad speed choice is kept

Button	Action
X / Square (hold)	The same aim + drive to the SHOT SPOT without firing; rumbles ON SPOT. Also switches MANUAL speed back to AUTO (camera)
Back / Share	Save the SHOT SPOT here (camera distance to the CELL in view, which CELL, current shot speed)
Y / Triangle	Flywheel pre-spin ON / OFF
D-pad UP / DOWN	MANUAL shot speed FAR (1570) / NEAR (1450) – see table above
D-pad RIGHT / LEFT	Shot speed +10 / −10 per press
LB + D-pad	Shot result for the log: UP IN, DOWN SHORT, RIGHT LONG, LEFT SIDE (see the D-pad table)
A / Cross	Flower arm DOWN / UP (only when stopped)
B / Circle (hold)	Unjam: windmill slowly backwards
INIT: B / A, Back, X, Y, L1, Options	Alliance RED / BLUE (remembered; AUTO sets it); diagnostics log on/off; aim gate on/off (off = FIRE never waits for the camera); auto shot speed on/off (off = NEAR / FAR only); field-centric on/off (OFF by default); Options = the intake points away from you now